Sprint Execution Plan (3 Sprints)

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Sprint 1: Architecture, Core MVP & Git Branching (Tasks 01 – 06)

• **Scope:** Project initialization, Spring Boot 3.2 skeleton, JPA entities, REST APIs, responsive web dashboard, GitFlow branching, Pull Requests, merge conflict resolution, and release baseline tagging v1.0.

• **Deliverables:** Functional local application running on port 8085 with 5 core MVP features and tagged Git history.

Sprint 2: Continuous Integration & Automated Testing (Tasks 07 – 10)

• **Scope:** Jenkins server setup, JDK 21 / Maven 3.9 configuration, Freestyle CI job with SCM polling, declarative Jenkinsfile pipeline, Selenium WebDriver test suite, quality gate defect injection, and recovery.

• **Deliverables:** Configured Jenkins pipeline, archived artifacts, Surefire JUnit XML reports, defect correction proof.

Sprint 3: Containerization & Ansible IaC (Tasks 11 – 15)

• **Scope:** Multi-stage Alpine JRE Dockerfile, container lifecycle testing, Jenkins-Docker continuous deployment, Ansible inventory and setup.yml playbook, idempotency proof, and final technical report compilation.

• **Deliverables:** Versioned Docker image, idempotent target provisioning, viva slide deck, and master documentation.